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orphan:

3.1 | Deck Design Properties

Import deck loads and functions.

Table 1: Load values from rv102-loads.py (rv_stor/v102-2.csv)

variable	value	[value]	description
D_1	2.00 plf	0.00 klf	2x6 planks DL
D_2	2.60 plf	0.00 klf	2x8 joists DL
D_3	2.90 plf	0.00 klf	4x4 posts and struts
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity
LL_1	40.00 psf	1.92 kPA	ASCE7-05 floor LL
HL_1	20.00 psf	0.96 kPA	ASCE7-05 HL

Table 2: Import Functions (rvsrc/scripts/checks.py)

Function	Docstring
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.
nds_post_check(** kwargs)	Check stress at cantilever post

3.1 - 2 | Deck Design Summary

Design properties as dictionary for checking function nds_beam_chk

```
Function Arguments Dictionary : beam1 (units: inch, pounds)
===========================================================================
ln_1 = 4*12.  # beam span
w_1 = 45*.5/12  # uniform linear load
b_1 = 5.5  # beam width
d_1 = 1.5  # beam depth
E_1 = 1.5*(10**6)  # modulus of elasticity
F_b = 1000 # allowable bending stress
C_D = 1.0  # load duration factor
C_M = 0.85 # wet service factor
C_F = 1.0  # size factor
C_t = 1.0  # temperature factor
C_i = 0.8  # incising factor
C_r = 1.0  # repetitive member factor
C_c = 1.0  # curvature factor
C_L = 1.0  # beam stability factor
C_b = 1.0  # bearing area factor
deflect_limit = 240.0 # max allowable deflection ln_1/deflect_limit
===========================================================================
```

Design Results

Eq. 1: Check Deck Beam

```
nds_beam_check | units: inch, pounds

     Beam Check Results:
     ===============================
     total UDL: 1.88
     fb: 261.82
     Fb_prime: 680.00
     E_prime: 1275000.00
     stress_ratio: 0.39
     deflection: 0.07
     deflection_ratio:  0.20
```

3.1 - 3 | Strut

Check strut D/C ratio with BeamChek 2023

BeamChek v2023 licensed to: StL Reg # 2999-68413

Tree Fort	Struts	
	Prepared by:	Date: 6/09/26

Selection — **4x 4 DF-L Select** **Solid Wood Column**

Conditions — NDS 2018, Using values for 2x and 4x solid sawn, Dimension Lumber.
Wet Use

Data

Load	1100 #	Column Area	12.25 in²	Kf	1.00
Actual Height	8.0 ft	le d1 Effective Ht	96 in	c	0.80
Unbraced L1	8.0 ft	le d2 Effective Ht	96 in	KcE	0.30
Unbraced L2	8.0 ft	Ke Buckling Mode	1.0	FcE	683

Attributes and Values — **Controlling d is 3.5 inches**

	le/d	psi	Area (in²)
Actual	27	90	12.25
Critical	50	606	1.82
Status	OK	OK	OK
Ratio	54%	15%	15%

	Fc ǁ (psi)	E (psi x mil)
Reference Values	1700	1.9
Adjusted Values	606	1.7
CF Size Factor	1.15	
Cd Duration	1.00	
Cm Wet Use	0.80	0.90
Cp Stability	0.39	

Note: A wood plate under this column must have an Fc value, perpendicular to the grain, greater than 90 psi.

Fig. 1 - Strut Check